

Failure Incident Report & RCA: Main Pump P-101

Investigation Date: May 14, 2026 | Inspector: Reliability Eng. Team

1. Incident Summary

On May 12, 2026, the main centrifugal feed pump P-101 tripped on bearing high-temperature limit (reaching 92 C). Operators reported loud metallic rattling and excessive discharge pressure fluctuations before the automatic trip. Flow rate had degraded to 40 GPM from the standard 120 GPM.

2. Failure Analysis & Findings

Post-incident inspection revealed the following physical details:

- Severe cavitation pitting was observed on the leading edges of the main impeller blades.
- The suction strainer S-101 was found completely clogged with paraffin debris and rust flakes from the tank lining, restricting incoming flow.
- High vibration levels (measured at 6.8 mm/s velocity) caused mechanical seal face misalignment, leading to leakage.
- The bearing oil level was found below 10%, indicating lubricant starvation which led to extreme friction heating.

3. Root Cause Identification

Primary Root Cause: Impeller cavitation caused by suction line fluid starvation, which originated from a clogged strainer. The bearing overheating was caused by bearing lubricant starvation due to missed lubrication cycles.

Contributing Factors: Lack of vibration transmitters, lack of automatic differential pressure alerts across strainer S-101, and manual lubrication routes not followed for two consecutive weeks.

4. Recommended Corrective Actions

- Clean suction strainer S-101 and implement a bi-weekly preventive cleaning schedule.
- Replace the damaged impeller blades and the mechanical seal face assembly.
- Install a digital differential pressure transmitter (DPT-101) across the strainer to alert control room of clog conditions.
- Revise and audit the manual lubrication route schedules to ensure bearing grease is topped up every 100 operating hours.